

MATH 1552 Midterm 1 Question Bank

Topic A

Riemann Sums and the Definition of the Definite Integral (5.1-5.3)

Name: _____

Date: _____

Instructions:

- Show all steps clearly and write final answers in simplified form.
- If you use a theorem (like the FTC), state which one you are using.
- For area problems, sketching the region is recommended.

Topic A: Riemann Sums and the Definition of the Definite Integral (5.1-5.3)

1. [15 points] Using the general form of the definite integral,

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x,$$

and choosing x_i^* as the **right-hand endpoint** of each subinterval, evaluate:

$$\int_1^3 (x^2 + 3x) dx.$$

Do not use the FTC.

2. [15 points] Using the general Riemann sum method,

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x,$$

evaluate the definite integral below. Assume n equally spaced subintervals and choose x_i^* as the right-hand endpoint.

$$\int_0^4 (2x^2 + 3) dx.$$

Any other method receives no credit.

3. [15 points] Using the general form of the definite integral,

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x,$$

and choosing x_i^* as the **right-hand endpoint** of each subinterval, evaluate:

$$\int_{-1}^1 (3x^2 + 4x + 2) dx.$$

Do not use the FTC.

4. [15 points] Use the method of Riemann sums (right-endpoint choice) to evaluate:

$$\int_{-1}^2 (6x^2 + 2x) dx.$$

Any other method receives no credit.

5. [15 points] Use the method of Riemann sums and the provided steps to evaluate:

$$\int_1^2 (3x - x^2) dx.$$

- () [4 points] Find Δx and an expression for x_i^* (using the numbers in this problem).

$$\Delta x = \underline{\hspace{2cm}} \qquad x_i^* = \underline{\hspace{2cm}}$$

- () [4 points] Evaluate $f(x_i^*)$.

- () [6 points] Compute $\sum_{i=1}^n f(x_i^*)\Delta x$ and simplify until it is only in terms of n .

- () [4 points] Take the limit as $n \rightarrow \infty$.

- () [9 points] Check by evaluating the integral directly using the FTC.

6. [30 points] Consider the function $f(x) = 1 + 6x$.

- () [10 points] Estimate the area bounded between the graph of f , the lines $x = -1$ and $x = 2$, and the x -axis using a **lower sum** with $n = 3$ equally spaced subintervals.

- () [14 points] Find the **actual area** in part (a) by taking the limit of a general Riemann sum using n equally spaced subintervals and choosing x_i^* to be the right-hand endpoint. **Any other method receives no credit.**

- () [6 points] Does your answer in part (b) represent a **total** area or a **net** area? Explain.

7. [23 points] Let $f(x) = x^2 + 2$.

- () [6 points] Find the value of the **lower Riemann sum** on $[-1, 2]$ using $n = 3$ evenly-spaced rectangles.

- () [4 points] Partition $[-1, 2]$ into n evenly-spaced subintervals. Find a formula for x_i , the right-hand endpoint of each subinterval.

- () [4 points] Using your answer from (b), find a formula for $f(x_i)$.

- () [9 points] Using your answers from (b) and (c), evaluate $\int_{-1}^2 f(x) dx$ using Riemann sums. You may choose x_i^* to be the right-hand endpoint. **Any other method receives no credit.**

8. **[16 points]** Use the limit definition of the definite integral to find the **average value** of

$$f(x) = (x + 2)^2$$

on the interval $[0, 3]$. You may choose right endpoints. **Any other method receives no credit.**